

Sana Layouni

Software Engineer – AI & Full-Stack Development | End-of-Studies Internship (PFE)

M'saken, Sousse, Tunisia • sanalayouni20@gmail.com • linkedin.com/in/sana-layouni • github.com/sanalayouni

SUMMARY

Software engineering student who builds web and mobile applications and enjoys turning AI into real products. Across 3 internships and several personal projects, I have shipped full-stack apps (React, Spring Boot, FastAPI, Flutter) and AI features powered by Large Language Models (LLMs), Retrieval-Augmented Generation (RAG) and semantic search. Microsoft Azure and NVIDIA certified, and a scholarship holder in an international computer science program in China. Seeking an end-of-studies internship (PFE) in software engineering and/or applied AI.

PROFESSIONAL EXPERIENCE

Axia Solutions, Sousse, Tunisia

Jul 2026 – Aug 2026

AI & Full-Stack Developer Intern

Tech: Python, LLMs, RAG, n8n, FastAPI, React, TypeScript, Flutter, PostgreSQL, Docker

- Designed and developed **HRHub**, an AI-powered HR platform that uses LLMs to classify employee email requests into 3 categories (leave, salary, remote work) and recommend policy-based decisions.
- Built a RAG pipeline with vector search (pgvector) over company policy documents, so that every recommendation is grounded in company policy (up from **43%** of runs before the redesign).
- Automated the end-to-end workflow with n8n: email ingestion, summarization, classification and routing to specialized AI agents.
- Guaranteed human-in-the-loop review on every request, with a confidence score and reasoning for each AI recommendation; used locally hosted LLMs so no employee data leaves the network.

Tagmanya

Jul 2025 – Sep 2025

Web Developer Intern

Tech: React, Node.js, Express, MongoDB, JWT, Git

- Developed *The Professional Cards*, a full-stack web application (MERN stack).
- Designed the MongoDB database schema and data models.
- Built REST APIs and secure user authentication with JSON Web Tokens (JWT).
- Created a responsive user interface for desktop and mobile.

DMnova

Jun 2024 – Jul 2024

Software Developer Intern

Tech: Unity, C#

- Developed a multiplayer quiz game with Unity and C#.
- Implemented the multiplayer flow, synchronizing questions, answers and scores between players.
- Built the core game logic and score management system.
- Designed a modular, object-oriented architecture to keep the code maintainable and easy to extend.

EDUCATION

Engineering Degree in Software Engineering – EPI-Digital School, Sousse, Tunisia

2024 – Present

International Computer Science Program (Academic Scholarship) – China

2025 – Present

Integrated Preparatory Cycle in Computer Science – ESPITA, Tunisia

2022 – 2024

Scientific Baccalaureate – Lycée Othman Chatti, M'saken, Tunisia

2022

SKILLS & OTHER

AI / ML: Machine Learning, Deep Learning, Natural Language Processing (NLP), LLMs, RAG, Prompt Engineering, AI Agents, Embeddings, Semantic Search, Hugging Face Transformers, LangChain, n8n

Programming: Python, Java, JavaScript, TypeScript, C#, Dart, SQL

Software Engineering: REST APIs, FastAPI, Spring Boot, Node.js, Express, React, Angular, Flutter, Object-Oriented Programming (OOP)

Data & Cloud: PostgreSQL, MongoDB, MySQL, Vector Databases (pgvector, ChromaDB, Qdrant), Microsoft Azure, Docker, Git, CI/CD, Agile / Scrum

Projects: **Job API** – semantic job-matching API (FastAPI, Sentence Transformers); compared 3 embedding models, reaching a **77.9%** top matching score. **Attention-RAG** – local RAG chatbot (LangChain, ChromaDB). **UtopiaHire** – AI job-search platform with job recommendation and skills analysis (React, Spring Boot).

Certifications: Microsoft Certified: Azure Fundamentals; NVIDIA – Building LLM Applications With Prompt Engineering; NVIDIA – Enhancing Data Science Outcomes With Efficient Workflow; NVIDIA – Introduction to Deep Learning; NVIDIA – AI for Anomaly Detection; Cisco – Python Basics

Languages: Arabic (native), French (advanced), English (advanced), German (beginner), Chinese (beginner)